

Week 4 - Data Quality

Power BI - The Insight Stack

What data quality does

View tab: Enable Column Quality, Column Distribution and Column Profile

Choose your profiling tool

Column Quality

Health check per column

- ✓ % Valid values
- ✓ % Error values
- ✓ % Empty values
- ✓ Quick column signal

Column Distribution

Value patterns and spread

- ✓ Distinct values
- ✓ Unique values
- ✓ Frequency patterns
- ✓ Dominant categories

Column Profile

Deep dive per column

- ✓ Min and max values
- ✓ Error and empty count
- ✓ Value distribution
- ✓ Column statistics

Key questions before profiling

Does the column have errors?

Are there missing values?

Are categories consistent?

Are data types correct?

Are there duplicates?

Is the column safe to use?

"Data preparation makes the data usable. Data quality checks make the data trustworthy."

Next up: Week 5 - Data Modelling

Why Data Quality Matters

Data quality is the process of checking whether data is reliable enough to support analysis and decision-making. Power Query provides built-in data profiling tools that help analysts inspect and understand the data before loading it into the model. Microsoft groups these tools into three main features: **Column quality**, **Column distribution**, and **Column profile**.

The key idea is simple: **Data preparation makes the data usable. Data quality checks make the data trustworthy.**

Data Quality Features Available in Power Query

Power Query provides three main profiling tools to help analysts assess data quality.

Power BI Data Profiling Features



Column Quality

Column quality shows the percentage of values that are valid, error, empty, unknown, or unexpected error within a column. It gives a quick visual signal of whether a column is safe to use or needs attention.



Column Distribution

Column distribution shows how values are spread across a column. It helps analysts identify frequent values, distinct values, unique values, repeated categories, and unusual distributions.



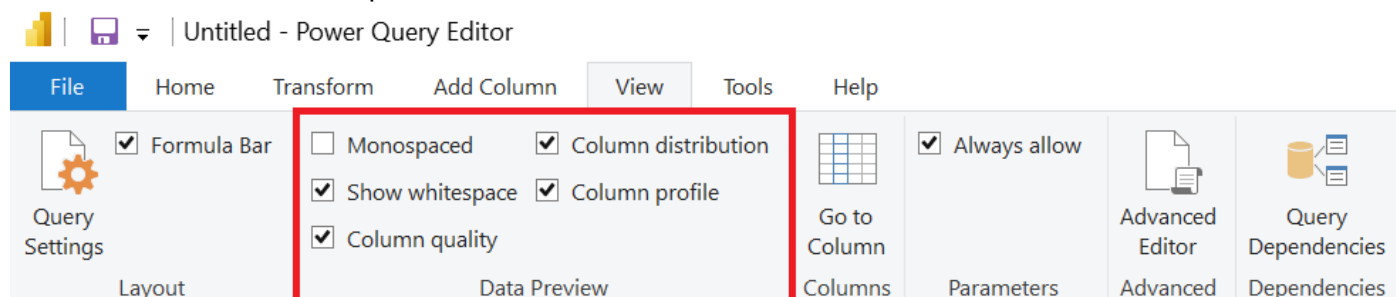
Column Profile

Column profile gives a deeper view of the selected column. It includes column statistics and value distribution, helping analysts inspect minimums, maximums, counts, blanks, distinct values, and other profiling details.

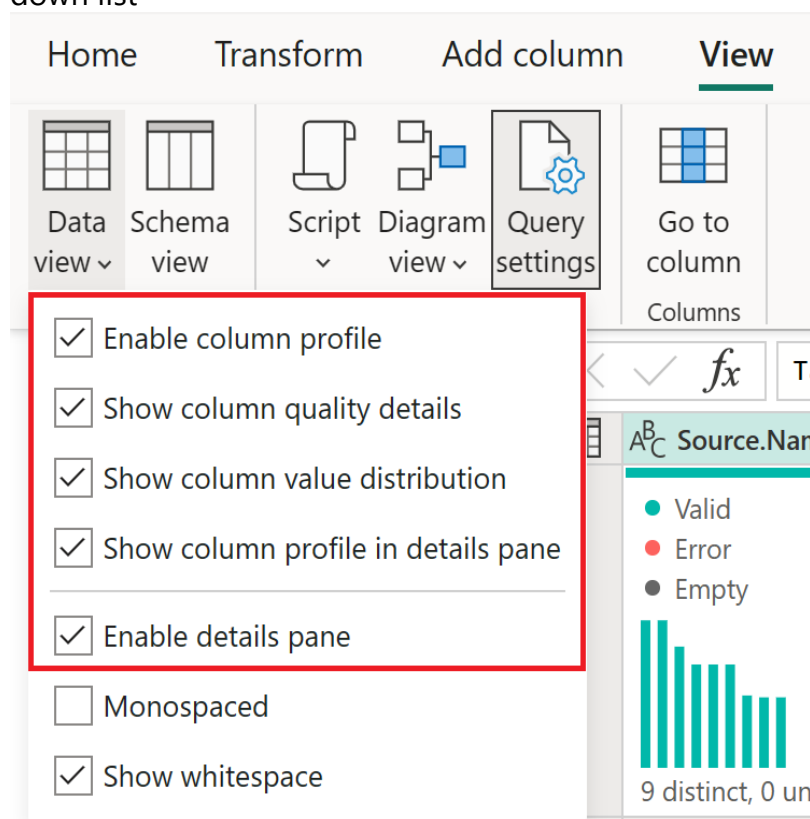
Together, these tools help analysts move from guessing about data quality to checking it directly inside Power Query.

How to Enable Data Profiling Tools

In **Power Query Desktop**, go to: View tab > Data Preview group > enable Column quality, Column distribution, and Column profile



In **Power Query Online**, go to: Data view > enable the required profiling options from the drop-down list

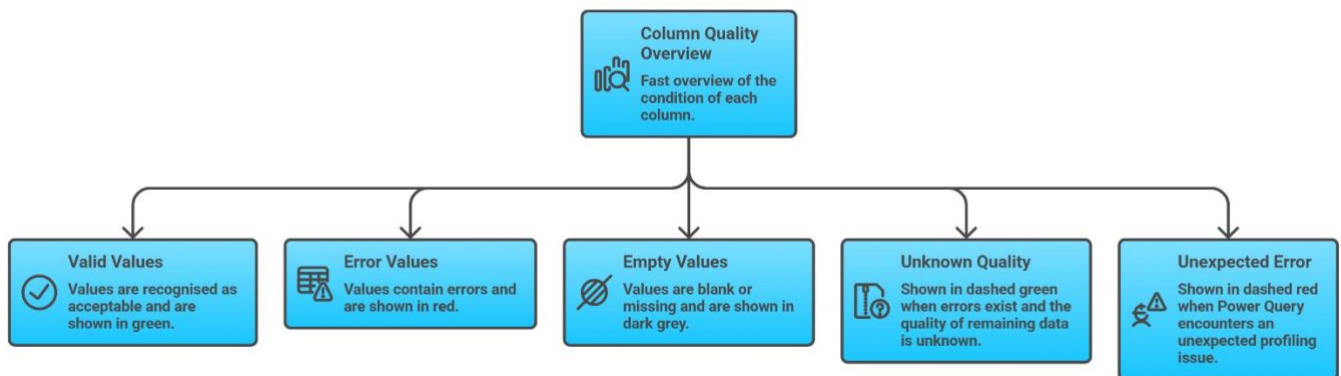


After enabling these options, the data profiling information appears inside the Power Query Editor. By default, Power Query profiles the first 1,000 rows, but analysts can change this to profile the entire dataset when needed.

Column Quality: Checking Valid, Error, and Empty Values

Column quality gives a fast overview of the condition of each column.

Column Quality Indicators



Made with Napkin

This is useful when analysts need to quickly answer:

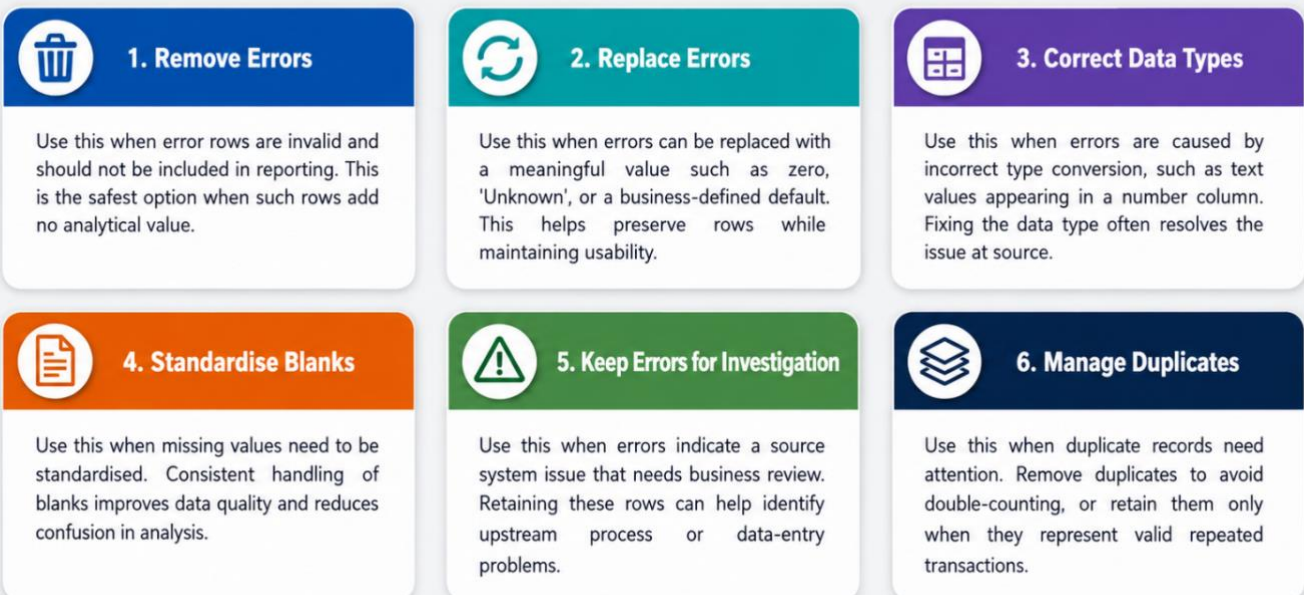
- Does this column contain errors?
- Are there missing values?
- Is the column safe to use in modelling?
- Should the column be cleaned before loading?

Column quality is often the first check because it gives a quick health signal across the dataset.

Error Handling: What to Do When Data Quality Issues Appear

Finding an error is not the end of the process. The analyst still needs to decide how to handle it.

Practical Error Handling Approaches



Power BI View to select each option:

	ABC	Name		\$	AnnualSales		\$
100%	Valid		100%	Valid		100%	
0%	Error		0%	Error		0%	
0%	Empty		0%	Empty		0%	
Name 701 (100%) 0 (0%) 0 (0%) Valid Error Empty distinct, 0 unique 5							
					800,000.00		
296		Riders Company					
298		The Bike Mechanics					
300		Nationwide Supply					
302		Area Bike Accessories					
304		Bicycle Accessories and Kits					
306		Clamps & Brackets Co.					
308		Valley Bicycle Specialists					
310		New Bikes Company			1,500,000.00		
312		Vinyl and Plastic Goods Corporation			1,500,000.00		

- Copy
- Keep Duplicates
- Keep Errors
- Remove Duplicates
- Remove Empty
- Remove Errors
- Replace Errors

Practice Activity Number 10

File

Home

Transform

Add Column

View

Tools

Help

Group By

Use First Row as Headers

Count Rows

Table

Transpose

Reverse Rows

Count Rows

Table

Data Type: Text

Detect Data Type

Rename

Any Column

Replace Values

Replace Values

Replace Errors

Any Column

Unpivot Columns

Move

Convert to List

Text Column

Split Column

Format

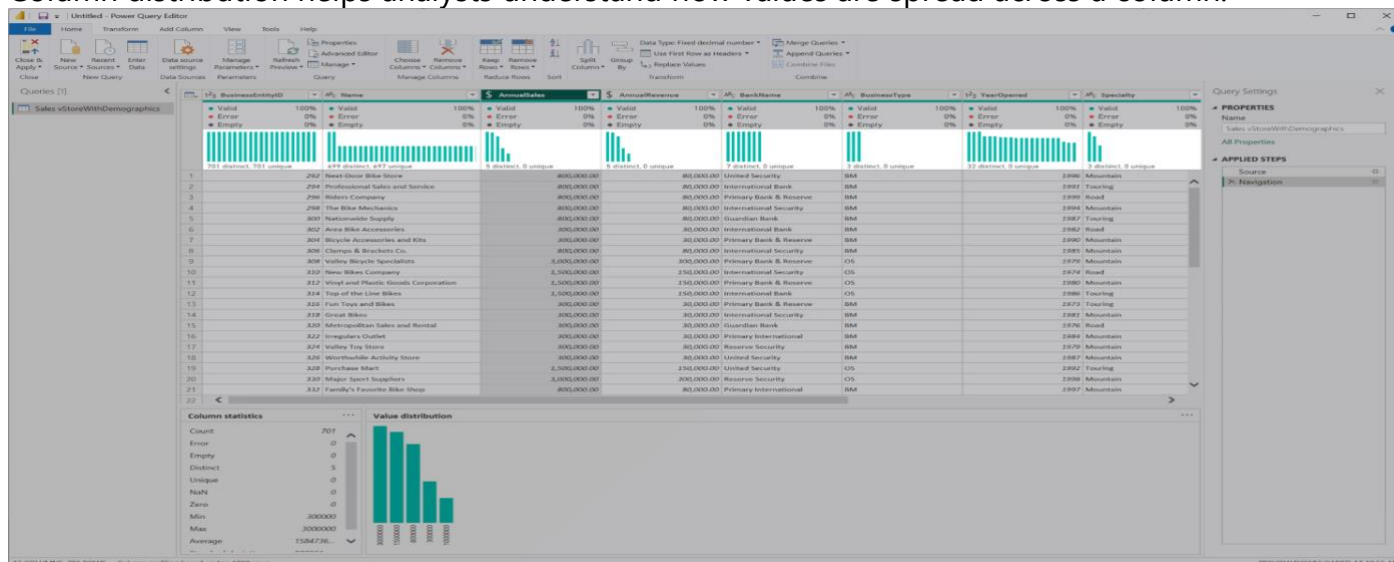
Extract

Parse

Text Column

Column Distribution: Understanding Value Patterns

Column distribution helps analysts understand how values are spread across a column.



It can help identify: Repeated values, Distinct values, Unique values, Dominant categories, Unusual patterns, Potential duplicates, Unexpected category names.

Microsoft explains that column distribution displays frequency and distribution visuals under each column, sorted from the most frequent value downward. It also shows distinct and unique values when analysts hover over the distribution.

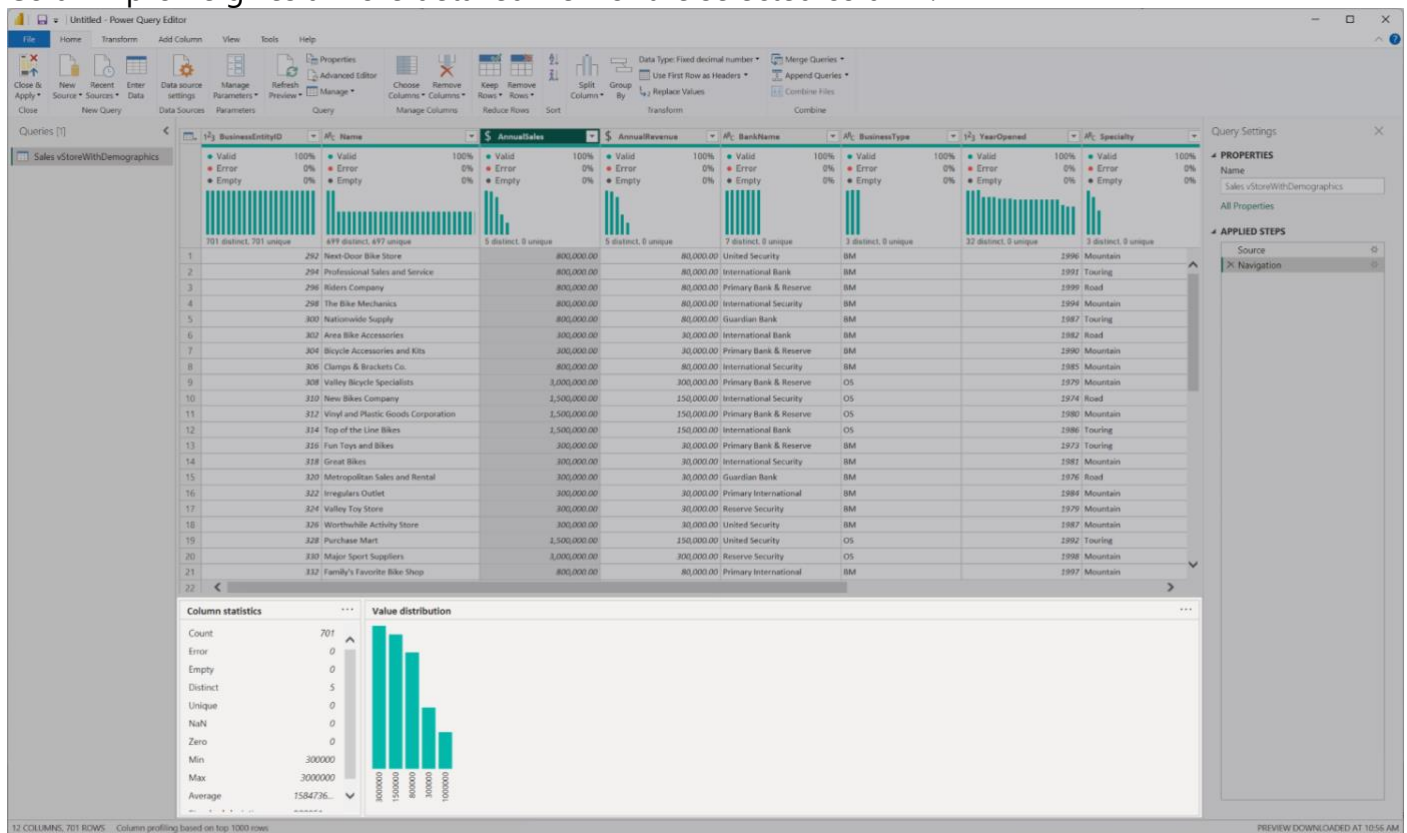
This is useful when asking:

- Are there too many unique values?
- Are categories repeated consistently?
- Are there spelling variations?
- Is one value dominating the column?
- Are there unexpected outliers?

For example, if a "Region" column contains both "Victoria", "VIC", and "Vic", column distribution can help reveal inconsistent category naming.

Column Profile: Deep Dive into a Column

Column profile gives a more detailed view of the selected column.



It includes: Column statistics, Value distribution, Count, Error count, Empty count, Distinct count, Unique count, Minimum and maximum values where relevant

This is useful for deeper investigation after column quality or column distribution highlights a concern.

For example:

- A date column may have unexpected minimum or maximum values.
- A product column may have too many unique values.
- A sales column may contain blanks or negative values.
- A category column may show unexpected groupings.

Column profile is best used when the analyst wants to inspect one column more carefully before deciding how to clean or transform it.

Practical Way to Decide: Which Profiling Tool Should You Use?



A Simple Decision Guide for Power Query Data Profiling

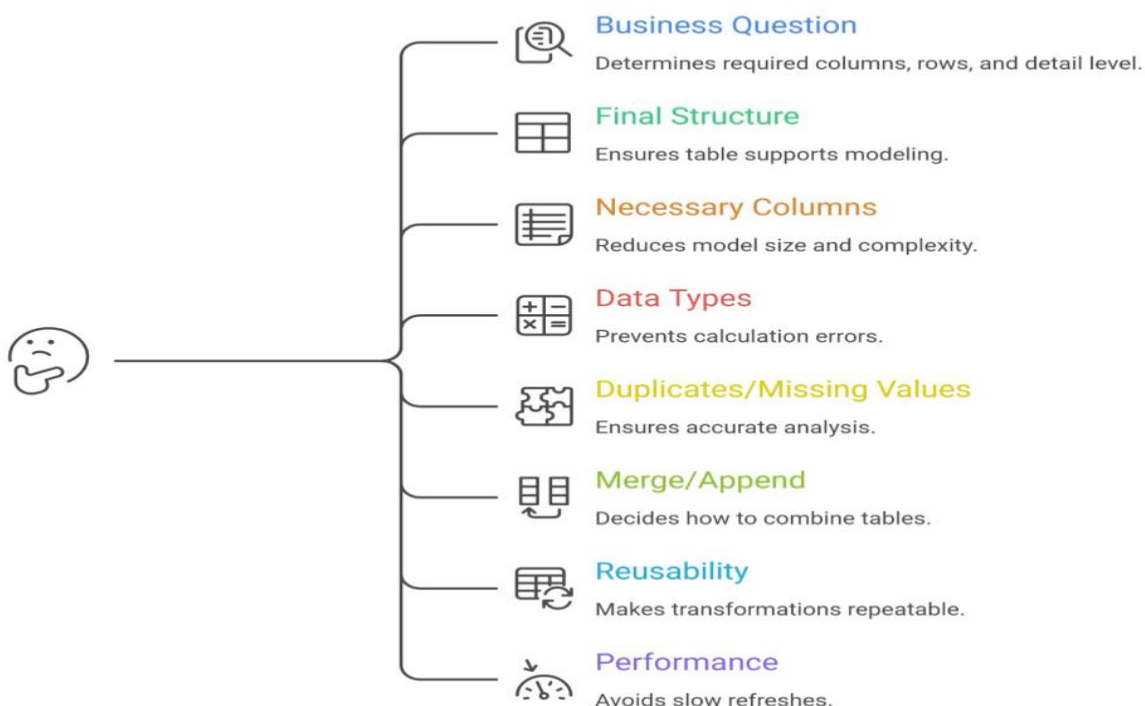
Choose the right profiling feature based on the data question you are solving.



Power Query profiling tools help turn raw data into analysis-ready tables.

What Analysts Should Ask Before Choosing Each Tool

What questions should be asked before cleaning data?



Conclusion

Data quality is the checkpoint between prepared data and trusted reporting. Power Query's profiling tools help analysts identify errors, blanks, repeated values, unusual distributions, and column-level issues before they affect the model or the final report.

The goal is not just to clean data. The goal is to understand whether the data is reliable enough to support the business decision.

Stay tuned for next week's article in **The Insight Stack: Power BI, Layer by Layer**, where we move into **Week 5: Data Modelling** and explore how analysts structure tables, relationships, and schemas to support reliable analysis.

Further Reading:

- Using the Data Profiling Tools: [Using the data profiling tools - Power Query | Microsoft Learn](#)
- Query Evaluation and Folding: [Understanding Query Evaluation and Query Folding in Power Query - Power Query | Microsoft Learn](#)